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JEE Main Physics Practice Sheet — DPP-035

Topic: Work Done by a Constant Force & Work as a Scalar (Dot) Product
Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

- Q1.** A constant force $\vec{F} = (3\hat{i} + 4\hat{j} - 5\hat{k})$ N displaces a body from position $\vec{r}_1 = (2\hat{i} - \hat{j} + 3\hat{k})$ m to $\vec{r}_2 = (5\hat{i} + 2\hat{j} - \hat{k})$ m. The work done by the force is:
- (A) 21 J
 (B) 41 J
 (C) 35 J
 (D) 15 J
- Q2.** A body of mass $m = 2$ kg is dropped from a height of $h = 20$ m. The work done by gravity on the body during its entire fall to the ground is ($g = 10$ m/s²):
- (A) -400 J
 (B) +400 J
 (C) +200 J
 (D) -200 J
- Q3.** A block of mass $m = 5$ kg is pulled along a horizontal floor by a constant force $F = 40$ N acting at an angle of 60° above the horizontal. If the block moves a distance of $s = 6$ m, the work done by the pulling force is:
- (A) 240 J
 (B) 60 J
 (C) 120 J
 (D) $120\sqrt{3}$ J
- Q4.** A porter carries a heavy suitcase of mass $M = 30$ kg on his head and walks a horizontal distance of $s = 50$ m on a level railway platform at a constant speed. The work done by the normal reaction from the porter's head on the suitcase is ($g = 10$ m/s²):
- (A) 15000 J
 (B) -15000 J
 (C) 1500 J
 (D) Zero
- Q5.** A body of mass $m = 4$ kg moves along the x -axis from $x = 0$ to $x = 5$ m under the action of a constant force $\vec{F} = (8\hat{i} - 6\hat{j})$ N. The work done by this force is:
- (A) 40 J
 (B) 10 J
 (C) 50 J
 (D) 70 J
- Q6.** A block of mass $m = 2$ kg slides down a rough incline of angle $\theta = 37^\circ$ and length $L = 5$ m at a constant velocity ($\mu_k = 0.75$). The work done by the frictional force during the descent is ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $g = 10$ m/s²):
- (A) +60 J
 (B) -60 J
 (C) -80 J
 (D) Zero
- Q7.** A particle moves in the xy -plane from $(0, 0)$ to $(3 \text{ m}, 4 \text{ m})$ under the simultaneous action of two constant forces $\vec{F}_1 = (2\hat{i} + 5\hat{j})$ N and $\vec{F}_2 = (4\hat{i} - 2\hat{j})$ N. The net work done on the particle is:
- (A) 26 J
 (B) 30 J
 (C) 45 J
 (D) 38 J
- Q8.** An object of mass m is tied to a string and whirled in a vertical circle of radius R . The work done by the tension force on the object during any quarter of a revolution is:
- (A) $\frac{1}{4}mgR$
 (B) $2\pi RT$
 (C) $\frac{1}{2}mv^2$
 (D) Zero
- Q9.** A constant force $\vec{F}$ acts on a particle moving with velocity $\vec{v}$. The work done per unit time (instantaneous power) is given by:
- (A) $\vec{F} \times \vec{v}$
 (B) $\frac{\vec{F}}{\vec{v}}$
 (C) $\vec{F} \cdot \vec{v}$
 (D) $\frac{1}{2}\vec{F} \cdot \vec{v}$
- Q10.** A body of mass $m = 10$ kg is pulled up a smooth inclined plane of angle $\theta = 30^\circ$ through a distance $s = 4$ m along the incline by a constant force $F = 100$ N parallel to the incline. The work done against gravity is ($g = 10$ m/s²):
- (A) 200 J
 (B) 400 J
 (C) $200\sqrt{3}$ J
 (D) 100 J
- Q11.** Under which of the following conditions is the work done by a non-zero force $\vec{F}$ on a moving body strictly negative?
- (A) Angle between $\vec{F}$ and displacement $\vec{s}$ is 0°
 (B) Angle between $\vec{F}$ and displacement $\vec{s}$ is acute ($0^\circ \leq \theta < 90^\circ$)

- (C) Angle between $\vec{F}$ and displacement $\vec{s}$ is obtuse ($90^\circ < \theta \leq 180^\circ$)
- (D) Angle between $\vec{F}$ and displacement $\vec{s}$ is exactly 90°
- Q12.** A constant force $\vec{F} = (a\hat{i} + 3\hat{j})$ N displaces a particle by $\vec{s} = (6\hat{i} - 4\hat{j})$ m. If the work done by the force is zero, the value of constant a is:
- (A) 4
(B) 2
(C) -2
(D) -4
- Q13.** A box is pushed across a horizontal surface by a force of 50 N directed at an angle of 37° below the horizontal. If the box is displaced by 4 m horizontally, the work done by the pushing force is ($\cos 37^\circ = 0.8$):
- (A) 120 J
(B) 200 J
(C) 160 J
(D) 80 J
- Q14.** A car of mass M accelerates uniformly from rest to speed V in time T on a level straight road. The work done by the engine during the first half of the time interval ($0 \leq t \leq T/2$) is:
- (A) $\frac{1}{8}MV^2$
(B) $\frac{1}{4}MV^2$
(C) $\frac{1}{2}MV^2$
(D) $\frac{3}{8}MV^2$
- Q15.** A pendulum bob of mass m suspended by a light string of length L is displaced through an angle θ_0 from the vertical and held. The work done by the external agent against gravity to displace it is:
- (A) $mgL \sin \theta_0$
(B) $mgL(1 - \cos \theta_0)$
(C) $mgL \cos \theta_0$
(D) $mgL(1 - \sin \theta_0)$
- Q16.** A body of mass $m = 3$ kg is projected vertically upward with speed $u = 20$ m/s. The work done by gravity during the time the body reaches its maximum height is ($g = 10$ m/s²):
- (A) -600 J
(B) +600 J
(C) -300 J
(D) +300 J
- Q17.** A force $\vec{F} = (2\hat{i} - 3\hat{j} + 4\hat{k})$ N acts on a body moving with constant velocity $\vec{v} = (3\hat{i} + 2\hat{j} + \hat{k})$ m/s. The rate of work done by the force is:
- (A) 16 W
(B) 4 W
(C) 10 W
(D) 0 W
- Q18.** A bucket filled with water of total mass 15 kg is pulled up from a well of depth 20 m with a uniform speed using a rope of linear mass density $\lambda = 0.5$ kg/m. The total work done against gravity in lifting the bucket and rope to the top is ($g = 10$ m/s²):
- (A) 3000 J
(B) 4000 J
(C) 3500 J
(D) 4500 J
- Q19.** A block of mass $m = 1$ kg is placed on the horizontal floor of an elevator ascending with an upward acceleration $a = 2$ m/s². The work done by the normal reaction on the block when the elevator moves up a distance $h = 5$ m is ($g = 10$ m/s²):
- (A) 50 J
(B) 10 J
(C) 40 J
(D) 60 J
- Q20.** A particle moves along the path $y = x^2$ from (0, 0) to (2 m, 4 m) under the action of a constant force $\vec{F} = (5\hat{i} + 3\hat{j})$ N. The work done by this force along the parabolic trajectory is:
- (A) 22 J
(B) 38 J
(C) 14 J
(D) 32 J

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-035)

Work Done by a Constant Force & Work as a Scalar (Dot) Product

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	C	16	A
2	B	7	B	12	B	17	B
3	C	8	D	13	C	18	B
4	D	9	C	14	A	19	D
5	A	10	A	15	B	20	A

Detailed Step-by-Step Solutions

Sol 1. Option (B)

Displacement vector:

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1 = (5-2)\hat{i} + (2-(-1))\hat{j} + (-1-3)\hat{k} = (3\hat{i} + 3\hat{j} - 4\hat{k}) \text{ m}$$

Work done:

$$W = \vec{F} \cdot \Delta \vec{r} = (3)(3) + (4)(3) + (-5)(-4) = 9 + 12 + 20 = 41 \text{ J}$$

Sol 2. Option (B)

Gravity acts downwards ($\vec{F}_g = -mg\hat{j}$) and displacement is downwards ($\vec{s} = -h\hat{j}$).

Since $\vec{F}_g$ and $\vec{s}$ are parallel ($\theta = 0^\circ$):

$$W_g = mgh = 2 \times 10 \times 20 = +400 \text{ J}$$

Sol 3. Option (C)

Work done by constant pulling force:

$$W = Fs \cos \theta = 40 \times 6 \times \cos 60^\circ = 240 \times 0.5 = 120 \text{ J}$$

Sol 4. Option (D)

The normal reaction acts vertically upwards ($\vec{N} = Mg\hat{j}$) while the displacement is purely horizontal ($\vec{s} = s\hat{i}$).

Since the angle between normal reaction and displacement is $\theta = 90^\circ$:

$$W = \vec{N} \cdot \vec{s} = Ns \cos 90^\circ = 0 \text{ J}$$

Sol 5. Option (A)

Displacement vector is $\vec{s} = 5\hat{i}$ m.

Work done:

$$W = \vec{F} \cdot \vec{s} = (8\hat{i} - 6\hat{j}) \cdot (5\hat{i}) = (8)(5) + (-6)(0) = 40 \text{ J}$$

Sol 6. Option (B)

Normal reaction on incline: $N = mg \cos 37^\circ = 2(10)(0.8) = 16 \text{ N}$.

Kinetic friction force: $f_k = \mu_k N = 0.75 \times 16 = 12 \text{ N}$.

Friction opposes motion along the incline ($\theta = 180^\circ$):

$$W_f = -f_k L = -12 \times 5 = -60 \text{ J}$$

Sol 7. Option (B)

Net force: $\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 = (2+4)\hat{i} + (5-2)\hat{j} = (6\hat{i} + 3\hat{j}) \text{ N}$.

Displacement: $\Delta \vec{r} = (3\hat{i} + 4\hat{j}) \text{ m}$.

Net work done:

$$W_{\text{net}} = \vec{F}_{\text{net}} \cdot \Delta \vec{r} = (6)(3) + (3)(4) = 18 + 12 = 30 \text{ J}$$

Sol 8. Option (D)

At every instantaneous point on the circular path, tension acts radially towards the center, whereas displacement is tangential ($\vec{T} \perp d\vec{r}$).

$$dW = \vec{T} \cdot d\vec{r} = 0 \implies W = 0$$

Sol 9. Option (C)

By definition:

$$P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$

Sol 10. Option (A)

Vertical height gained: $h = s \sin \theta = 4 \sin 30^\circ = 4(0.5) = 2 \text{ m}$.

Work done against gravity:

$$W_{\text{against } g} = mgh = 10 \times 10 \times 2 = 200 \text{ J}$$

Sol 11. Option (C)

$W = Fs \cos \theta$. For $90^\circ < \theta \leq 180^\circ$, $\cos \theta < 0$, making work done strictly negative.

Sol 12. Option (B)

$$W = \vec{F} \cdot \vec{s} = (a)(6) + (3)(-4) = 0 \implies 6a - 12 = 0 \implies a = 2.$$

Sol 13. Option (C)

Horizontal component of pushing force is $F_x = F \cos 37^\circ = 50(0.8) = 40 \text{ N}$.

Displacement is horizontal ($s = 4 \text{ m}$):

$$W = F_x \cdot s = 40 \times 4 = 160 \text{ J}$$

Sol 14. Option (A)

Uniform acceleration $a = V/T$.

At $t = T/2$, speed is $v = a(T/2) = \frac{V}{2}$.

By Work-Energy Theorem:

$$W = \Delta K = \frac{1}{2}Mv^2 - 0 = \frac{1}{2}M\left(\frac{V}{2}\right)^2 = \frac{1}{8}MV^2$$

Sol 15. Option (B)

Vertical height through which bob is raised: $h = L - L \cos \theta_0 = L(1 - \cos \theta_0)$.

Work done against gravity:

$$W = mgh = mgL(1 - \cos \theta_0)$$

Sol 16. Option (A)

Maximum height reached: $H = \frac{u^2}{2g} = \frac{20^2}{2(10)} = 20 \text{ m}$.

Force of gravity is downward ($-mg\hat{j}$) and displacement is upward ($+H\hat{j}$):

$$W_g = -mgH = -(3 \times 10 \times 20) = -600 \text{ J}$$

Sol 17. Option (B)

Instantaneous power / rate of work:

$$P = \vec{F} \cdot \vec{v} = (2)(3) + (-3)(2) + (4)(1) = 6 - 6 + 4 = 4 \text{ W}$$

Sol 18. Option (B)

Mass of rope: $M_{\text{rope}} = \lambda H = 0.5 \times 20 = 10 \text{ kg}$.

Work to lift bucket: $W_1 = M_{\text{bucket}}gH = 15 \times 10 \times 20 = 3000 \text{ J}$.

Work to lift rope (center of mass lifted by $H/2$):

$$W_2 = M_{\text{rope}}g \left(\frac{H}{2} \right) = 10 \times 10 \times 10 = 1000 \text{ J}$$

Total work: $W = W_1 + W_2 = 3000 + 1000 = 4000 \text{ J}$.

Sol 19. Option (D)

In the accelerating elevator, the normal reaction on the block is:

$$N = m(g + a) = 1(10 + 2) = 12 \text{ N}$$

Both normal reaction and displacement are vertically upward ($\theta = 0^\circ$):

$$W_N = Nh = 12 \times 5 = 60 \text{ J}$$

Sol 20. Option (A)

Since force $\vec{F} = (5\hat{i} + 3\hat{j}) \text{ N}$ is constant, work done depends only on the initial and final position coordinates:

$$\Delta\vec{r} = (2 - 0)\hat{i} + (4 - 0)\hat{j} = (2\hat{i} + 4\hat{j}) \text{ m}$$

$$W = \vec{F} \cdot \Delta\vec{r} = (5)(2) + (3)(4) = 10 + 12 = 22 \text{ J}$$